

MOHAMED MAHDY

Full-Stack Developer | Python · Django/DRF · React · PostgreSQL · AI/RAG Systems | Cairo, Egypt

mohamedmahdy648@gmail.com | +20 112 854 8773 | linkedin.com/in/mohamed-mahdy257 | github.com/Mohamed-Mahdy7

SUMMARY

Full-stack developer (Python/Django/React) with 2+ year of production backend experience and a Business Administration background that sharpens stakeholder communication. Tech-led a 4-engineer team shipping AcademiQ, a Django/DRF + React + PostgreSQL SaaS platform, personally architecting its RAG/AI layer (Gemini API, pgvector, Celery) and 47 documented REST endpoints. Built and maintained production APIs, JWT auth, and Odoo 17 ERP customizations for real business clients at Trust Grp. Targeting Backend / Full-Stack Engineer roles where I can own systems end to end.

SKILLS

Languages: Python, JavaScript (ES6+), SQL, Bash

Backend: Django, Django REST Framework (DRF), Flask, SQLAlchemy, Celery, REST API Design, JWT Auth, RBAC, WebSockets

Frontend: React, HTML5, CSS3, Bootstrap, jQuery, TailwindCSS

Databases: PostgreSQL, pgvector, MySQL, Microsoft SQL Server, MongoDB

AI / RAG Engineering: RAG pipeline design, pgvector, Gemini API, LLM integration, OpenAI fine-tuning, prompt engineering, AI cost control & usage logging

Cloud & DevOps: Docker, Railway, Vercel, Git/GitHub, Linux, Redis, Postman, CI/CD basics

ERP: Odoo 17 development, financial/accounting reporting (PFA), Arabic localization

Practices: Agile/Scrum, sprint planning, ERD & API documentation, unit testing, code review, technical leadership

EXPERIENCE

Tech Lead & Full-Stack / AI Developer — AcademiQ (ITI Graduation Project)

Feb 2026 – Jun 2026

- Tech-led a 4-engineer team as sole architect: owned all technical decisions, ran sprint planning, and broke scope into 58 tickets across 6 sprints, shipping a production-ready Django/DRF + React + PostgreSQL SaaS platform in 4 weeks.
- Architected and built the platform's RAG pipeline from scratch (pgvector + Gemini API), powering automated report-card generation.
- Engineered a Celery-based dropout-risk detection engine that scans student activity on a schedule and auto-flags at-risk students, replacing a fully manual review process and cutting admin follow-up time by 15%.
- Built an AI cost-control layer (response caching + usage logging) that reduced redundant LLM calls.
- Delivered 47 documented REST API endpoints with RBAC and multilingual support; defined a single interface contract (`get_student_context()`) so 3 teammates could consume AI features without touching model internals, eliminating integration bugs from direct DB access.
- Deployed backend to Railway and frontend to Vercel, running 2 releases with zero downtime.

Full-Stack Developer — Trust Grp

Nov 2024 – Dec 2025

- Built and shipped Python/Flask backend services powering core business workflows, integrating with frontend to deliver end-to-end features for users.
- Designed and documented RESTful APIs for third-party integrations, reducing partner integration time through clear API docs and Postman collections.
- Implemented JWT-based authentication and authorization across 3 services, closing and securing every protected endpoint.
- Tuned SQL Server queries, indexes, and schema design.
- Owned production deployments and database migrations across 2 releases with zero data-loss incidents.

Odoo Developer — Trust Grp

Jul 2024 – Oct 2024

- Customized Python-based Odoo 17 modules covering 4 business processes, including Arabic-localized financial reporting used by end users.
- Optimized PostgreSQL queries powering ERP workflows.
- Automated recurring backend processes via scheduled Python jobs.

Trainee — Full-Stack & Generative AI Development — Information Technology Institute (ITI) *Jan 2026 – Jun 2026*

- Completed a 6-month, full-time Full-Stack Web & Generative AI Development track (Python): frontend (HTML/CSS/JS/React), backend (Django/DRF/Flask), databases (PostgreSQL/MySQL), and applied LLM/RAG integration — capstone project: AcademiQ (above).

PROJECTS

AcademiQ — Full-Stack SaaS + RAG/AI Platform | Jun 2026 — [GitHub](#)

- Tech-led a 4-engineer team building a Django/DRF + React + PostgreSQL SaaS platform end to end; personally built the AI infrastructure (Gemini client, RAG engine, pgvector, Celery automation).

PDFs Chat — RAG Platform | Sep 2025 — [GitHub](#)

- Built a Django-based RAG platform with authentication, RBAC, and WebSocket-driven real-time chat responses.

Vegist — E-Commerce Web App | Feb 2026 – Mar 2026 — [GitHub](#)

- Built a multi-actor e-commerce application (HTML5, Bootstrap 5, JavaScript, jQuery) with LocalStorage-based persistence.

Student Grade Management System (SGMS) — Bash CLI | Apr 2026 — [GitHub](#)

- Built a CLI grade-management tool with flat-file storage, full CRUD, and GPA calculations, validated by a 240-case automated test suite (238/240 passing, 99% coverage).

Spaxet — POS & Session Management | Nov 2024 – Dec 2025

- Built a Flask backend on Microsoft SQL Server (SQLAlchemy/pyodbc) exposing RESTful APIs for point-of-sale and session management, with third-party integration support.

Finvo — Client & Invoice Management

- Built a Flask + SQL Server application for client and invoice management using SQLAlchemy and pyodbc.

EDUCATION

Bachelor's Degree in Business Administration

Faculty of Commerce, Tanta University — 2022

CERTIFICATIONS

- McKinsey Forward Program — McKinsey.org (Dec 2025) — [Certificate](#)
- CS50x: Introduction to Computer Science — Harvard University (Dec 2025) — [Certificate](#)